

Elastic Load Balancer (ELB) and Elastic Block Storage (EBS)

To manage and distribute traffic across multiple servers, we use an Elastic Load Balancer (ELB). The ELB balances traffic and ensures high availability by routing requests to healthy servers. If a server goes down, the ELB redirects traffic to another available server.

Servers are grouped into what's called a target group. When users access a website, the traffic first reaches the load balancer, which then distributes it to the active servers within the target group. If a server in the target group is unhealthy, the load balancer redirects the traffic to another healthy server.

Setting Up ELB and Target Groups

1. **Launch a group of servers.**

Connect to each server and install the required package, such as the Apache HTTP server (`httpd`), using the following command:

bash

Copy code

```
sudo yum install httpd -y
```

- 2.

Start the HTTP service with:

bash

Copy code

```
sudo systemctl start httpd
```

- 3.

Check the service status to ensure it's running:

bash

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```
sudo systemctl status httpd
```

- 4.

Enable the service to start on boot:

bash

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```
sudo systemctl enable httpd
```

- 5.

6. The default web server path is `/var/www/html/`. You can place your website files here.

Creating Target Groups

1. Go to the **Target Groups** section under the Load Balancer in AWS.
2. **Create a target group** and specify the protocol and port.
3. **Configure health checks**, providing the path, timeout, healthy and unhealthy thresholds, and success codes.
4. Optionally, add tags and proceed to the next step.
5. **Register your servers** in the target group, selecting the servers and including them as pending.
6. **Create the target group**.

Creating a Load Balancer

AWS provides four types of load balancers:

1. **Application Load Balancer**: Works at the HTTP/HTTPS layer (Layer 7).
2. **Network Load Balancer**: Works with TCP/UDP (Layer 4).
3. **Gateway Load Balancer**: Routes traffic based on specific conditions.
4. **Classic Load Balancer**: The older version that combines features of both Layer 4 and Layer 7.

Understanding Load Balancer Layers

To effectively use ELB, it's important to understand the OSI model, particularly the layers relevant to your chosen load balancer. For example, an Application Load Balancer operates at Layer 7.

Configuring a Load Balancer

1. **Create the load balancer** and configure it according to your requirements.
2. Each load balancer has an ARN (Amazon Resource Name) and a DNS name.
3. **Copy the DNS URL** and paste it into your web browser to access the application.

Troubleshooting and Monitoring

- If the service stops, you might get a 502 error, indicating the server is not working.
- Other issues, like networking or mapping problems, may cause errors.
- Check your target groups to see how many servers are active or inactive.
- If a server is deregistered from a target group, it is considered "draining." Unhealthy servers can be identified based on error codes.

Monitoring and Scaling

AWS provides CloudWatch for monitoring ELB. Additionally, Auto Scaling is available to automatically create new servers based on traffic flow, ensuring the application remains responsive under varying loads.